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MILITARY SPECIFICATION

BOOTS, FLYERS' FWU-8/P

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of quick-donning, lined, flight boots which are designated FWU-8/P.

* 1.2 Classification. The boots shall be furnished in the sizes and the widths listed in table I as specified (see 6.2).

TABLE I. Schedule of Sizes

Sizes	Widths						
	A	B	C	D	E	EE	EEE
6-1/2	X	X	X	X	X		
7	X	X	X	X	X		
7-1/2	X	X	X	X	X	X	
8	X	X	X	X	X	X	
8-1/2	X	X	X	X	X	X	
9	X	X	X	X	X	X	
9-1/2	X	X	X	X	X	X	
10	X	X	X	X	X	X	
10-1/2	X	X	X	X	X	X	
11	X	X	X	X	X	X	X
11-1/2	X	X	X	X	X	X	X
12		X	X	X	X	X	X
12-1/2		X	X	X	X		

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

V-L-61 Laces, Nylon
V-T-276 Thread, Cotton

FSC 8430

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V-T-285	Thread, Polyester
V-T-291	Thread, Linen
V-T-295	Thread, Nylon
KK-I-570	Insole, Footwear, Leather, Cattlehide
TT-S-735	Standard Test Fluids, Hydrocarbon
UU-P-268	Paper, Kraft, Wrapping
CCC-C-443	Cloth, Duck, Cotton (Single and Plied Filling Yarns, Flat)
DDD-T-86	Tape, Textile; Cotton, General Purpose (Unbleached, Bleached, or Dyed)
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing

Military

MIL-S-4383	Sealing Compound, Topcoat, Fuel Tank, Buna-N Type
MIL-L-10867	Leather, Cattlehide, Gusset, Chrome Tanned, Fatliquored
MIL-B-17757	Boxes, Fiberboard, Corrugated (Modular Sizes)
MIL-F-21840	Fastener Tapes, Hook and Pile, Synthetic
MIL-S-22777	Soles and Heels, Rubber, Traction Tread, Shoe
MIL-C-41814	Counter, Footwear

STANDARDS

Federal

FED-STD-191	Textile Test Methods
FED-STD-311	Leather, Methods of Sampling and Testing
FED-STD-601	Rubber: Sampling and Testing
FED-STD-751	Stitches, Seams, and Stitchings

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

DRAWING

Air Force

73186	Clamp Assembly, Clinch-up, Boot, Flyer's FWU-8/P

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

B 36	Brass Plate, Sheet, Strip, and Rolled Bar
B 134	Brass, Wire
D 5	Penetration of Bituminous Materials
D 412	Tension Testing of Vulcanized Rubber
D 570	Water Absorption of Plastics
D 746	Brittleness Temperature of Plastics and Elastomers by Impact
D 792	Specific Gravity and Density of Plastics by Displacement
D 816	Rubber Cements
D 1043	Stiffness Properties of Plastics as a Function of Temperature by Means of a Torsion Test
D 1052	Measurement of Cut Growth of Rubber by the Use of the Ross Flexing Machine
D 1084	Viscosity of Adhesives
D 1203	Loss of Plasticizer from Plastics (Activiated Carbon Methods)
D 1238	Measuring Flow Rates of Thermoplastics by Extrusion Plastometer
D 2240	Indentation Hardness of Rubber and Plastics by Means of a Durometer
D 2941	Measuring Break Pattern of Leather (Break Scale)
E 18	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Metals
E 28	Softening Point by Ring-and-Ball Apparatus

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries and are also distributed among technical groups and using federal agencies.)

3. REQUIREMENTS

* 3.1 First article inspection. When specified by the procuring activity (see 4.3 and 6.2), this specification provides for first article inspection.

3.2 Guide samples. Samples of the boots, if furnished (see 6.3), are solely for information and guidance. In the event the samples deviate from the requirements of this specification, the requirements of this specification shall govern.

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3.3 Materials

* 3.3.1 Upper leather. The leather for the vamps, the quarters, and the back stays shall be cut from best quality, green salted, chrome-tanned sides. The sides shall be full grained leather and shall be 3-1/4 to 4-1/2 ounces thick. The area of each side shall not exceed 23 square feet. The tannery lot shall not exceed 25,000 square feet. The leather shall be similar in softness to elk-type sides and shall have a break pattern of not more than 5-grade rating (see ASTM D 2941). Cut leather parts that contain scratches, slaughter cuts on the grain surface of the leather, brands, or light flesh cuts that show through on the grain surface shall not be used. Leather defects such as fat wrinkles, pit, and insect damage plainly visible shall not appear in the vamps or quarters in front of the breast of the heel. Clusters of veins, split veins, or pronounced veins shall not appear in the vamps or the quarters. Flanky, pipey, loose, cracked, or any other rough or coarse grain leather shall not appear in the counter area back of the breast of the heel or in the tongue or backstay.

* 3.3.1.1 Color and finish. The leather shall be treated with a suitable water resistant compound, shall be drum dyed, and shall be black. The leather shall be full grain and shall be finished grain side out with no buffing or snuffing to remove surface blemishes. A light application of black finish containing a sufficient amount of pigment to obtain a uniform color shall be applied to the surface of the leather, and the surface of the leather shall be brought to a smooth finish. The flesh side of the leather shall be cleaned of coarse fibers. A fungicide paranitrophenol treatment of the leather shall not be required.

* 3.3.2 Lining. The leather for the quarter lining, the vamp lining, and the gusset lining shall be soft, chrome-tanned, full grain or corrected grain cattlehide glove leather that is 2 to 3-1/2 ounces thick and shall not require a fungicide paranitrophenol treatment. The color of the leather for the lining shall be cream or light russet. The leather for the lining shall have a smooth finish but shall not have a painted finish.

* 3.3.3 Gusset and top cuff. The gusset and the top cuff shall be black leather conforming to treatment C of MIL-L-10867.

3.3.4 Insoles. The insoles shall conform to type I, class 1, tannage a or b of KK-I-570 and shall be not less than 6 irons nor more than 7 irons thick.

* 3.3.5 Heel pads. The heel pads shall be cut from the leather specified in 3.3.2, sheepskin, natural kip, or calf. The sheepskin, natural kip, or calf shall weigh not less than 2 ounces nor more than 3-1/2 ounces.

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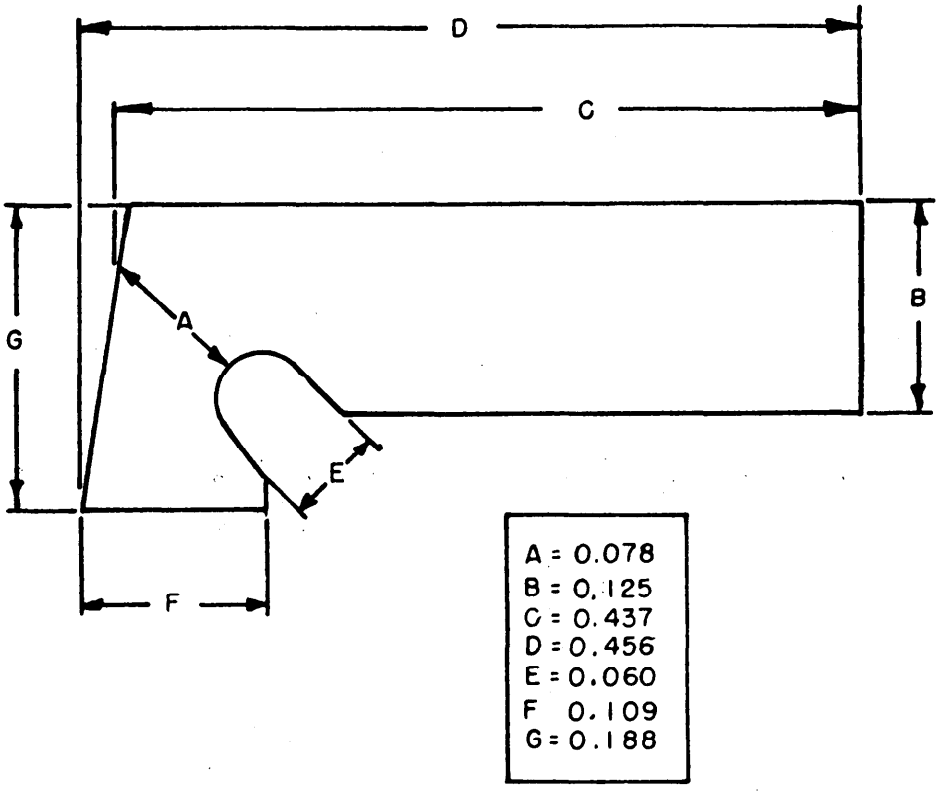
* 3.3.6 Welting. The welting shall be black, bright, flat top conforming to the dimensions shown on figure 1. The base compound shall be a virgin polyvinyl chloride compound with suitable nonbleeding plasticizers. The finished welting shall be smooth; uniform in texture; and free from porosity, blemishes, cracks, blisters, or any other defects that might affect serviceability or appearance. The finished welting shall conform to the requirements specified in table II.

* TABLE II. Welting

Property	Requirement
Hardness	85 - 95
Specific gravity	1.32 (maximum)
Brittle point	
Before aging	Minus 15 degrees Fahrenheit or less
After aging	Plus 20 degrees Fahrenheit change from standard brittle point (maximum)
Tensile strength	2,300 pounds per square inch (psi) (minimum)
Elongation	225 percent (minimum)
Water absorption	2.2 percent gain (maximum)
Solvent resistance	
Hardness	60 (minimum)
Tensile strength	1,800 psi (minimum)
Elongation	225 percent (minimum)
Volume swell	30 percent (maximum)

* 3.3.7 Counters. The counters shall be either leather counters or polyethylene counters.

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UNLESS OTHERWISE SPECIFIED, DIMENSIONS IN INCHES AND TOLERANCES ±2 PERCENT

FIGURE 1. Die for Welt

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* 3.3.7.1 Leather counters. The leather counters shall be welt type, full molded counters that conform to MIL-C-41814 except that the counters shall be 5 to 6 irons thick and shall be shaped to fit the Type FWU-5/P last.

* 3.3.7.2 Polyethylene counters. The polyethylene counters shall be molded, to fit the last, from natural-color, unfilled polyethylene resin that conforms to table III. The molded pieces shall be uniform in texture and finish and shall be free from porosity, cracks, blisters, and any other defects that will affect serviceability.

* TABLE III. Polyethylene Resin

Characteristic	Requirement
Specific gravity	0.91 to 0.93
Melt index	1.5 to 4.0
Tensile strength	1,400 to 2,500 psi
Elongation	100 to 800 percent
Stiffness modulus (torsion) to 23 degrees centigrade	20,000 to 40,000 psi
Brittle temperature (80 percent nonfailure)	Minus 76 degrees Fahrenheit (minimum)
Water absorption, gain in weight in 24 hours	0.14 percent (maximum)
Volatility loss (72 hours at 82 degrees centigrade)	0.05 percent (maximum)

* 3.3.8 Heels. The heels shall conform to type II, grade A, style 2 of MIL-S-22777.

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* 3.3.9 Outersoles. The outersoles shall conform to type I, grade A, style 2 of MIL-S-22777.

* 3.3.10 Midsoles. The midsoles shall be made from a black, jet fuel resistant, rubber compound and shall conform to the requirements specified in table IV.

* TABLE IV. Midsoles

Property	Requirement
Composition	Nonmarking
Thickness	3 irons
Hardness	88 (minimum) 95 (maximum)
Cut growth after aging and 25,000 flexes	500 percent (maximum)
Volume swell	30 percent (maximum)

* 3.3.11 Insole reinforcement. The material for the insole reinforcement shall conform to type I or II of CCC-C-443.

*

3.3.12 Tape. The tape for reinforcing the insoles, if used, and the tape for the back seam shall conform to type I, class 1 of DDD-T-86 except that the requirements for nonfibrous materials shall not apply. The width of the tape shall be 3/4 inch for reinforcing the insoles and 1/2 inch for back seams.

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3.3.13 Thread

* 3.3.13.1 Upper stitching operations. Unless otherwise specified herein, the needle thread for the upper stitching operations shall be dyed black to match the color of the upper leather and shall conform to type IB3, 4 ply, ticket No. 16 of V-T-276 except that the colorfastness requirements shall not apply. Unless otherwise specified herein, the bobbin thread for the upper stitching operations shall conform to type IA1 or IA3, 4 ply, ticket No. 16 of V-T-276 except that the type IA3 thread shall be black and the colorfastness requirements shall not apply. As an alternate to the cotton thread, the needle thread and the bobbin thread for the upper stitching operations may be nylon thread conforming to type I or II, class 1 of V-T-295 or to type III of V-T-295.

* 3.3.13.2 Goodyear stitching. The running thread for the Goodyear stitching shall be dyed to match the color of the upper leather. The shuttle thread for the Goodyear stitching operations shall be unbleached (natural color). Unless otherwise specified herein, the running thread for the Goodyear stitching operations shall be cotton thread conforming to type IVB, ticket No. 8, 11 ply, Z twist of V-T-276 except that the colorfastness requirements of V-T-276 shall not apply. Unless otherwise specified herein, the shuttle thread for the Goodyear stitching shall be cotton thread conforming to type IVA, ticket No. 8, 10 ply of V-T-276 except that the colorfastness requirements of V-T-276 shall not apply. The cotton thread for the Goodyear stitching shall be well impregnated with wax. As an alternate to the cotton thread for the Goodyear stitching, the running thread and the shuttle thread for the Goodyear stitching may be a polyester thread conforming to type I, class 1, size 10, 3 or 6 ply of V-T-285.

* 3.3.13.3 Inseaming operations. Unless otherwise specified herein, the thread for the inseaming operations shall be linen thread conforming to type B, class 1, 10 ply, right twist of V-T-291 except that the linen thread shall be water resistant and shall have a minimum breaking strength of 55 pounds. The linen thread for the inseaming operations shall be well impregnated with wax. As an alternate to the linen thread, the thread for the inseaming operations may be a polyester thread conforming to type I, class 1, size 10, 3 or 6 ply of V-T-285.

* 3.3.14 Fastener tape. The fastener tape for the top cuff and the top of the boot shall be 3/4 inch wide, a minimum of 2 inches long, and black. The pile fastener tape shall conform to class 1 of MIL-F-21840. The hook fastener tape shall conform to type I, class 1 of MIL-F-21840.

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* 3.3.15 Plastic box toes. The material for the box toes shall be a woven flannel or a processed (needled) fabric base impregnated with a pyroxylin or polystyrene compound and shall be 0.0875 inch (plus or minus 0.0075 inch) thick. The box toes shall be either an activated or a prepared type. The box toes shall conform in shape and size to the Type FWU-5/P lasts. The box toes shall be skived 9/16 inch (plus or minus 1/16 inch) with a feather scarf edge.

* 3.3.16 Cinch clamp. The cinch clamp (see 6.8) shall be aluminum and shall conform to Drawing 73186.

* 3.3.17 Speed lacers. The speed lacers shall conform to Style 31092 (manufactured by J. L. Thomson Fastener and Machine, Inc., Franklin Park, Illinois), or equal. The shaft on the speed lacers shall be of sufficient length to make a secure clinch. After fabrication, the speed lacers (speed clip and rivet) shall be anodized black or tumble coated with not less than two coats of black enamel.

* 3.3.17.1 Eyelets. The two eyelets (one on each blucher nose) required for each boot shall be made from aluminum that is not less than 0.012 inch thick and shall have a roll setting barrel. The eyelets shall be anodized or finished with a corrosion-preventive treatment. The heads of the eyelets shall be roller coated with two coats of enamel in a color that will approximately match the color of the upper leather. The eyelets may be tumble (fully) coated, with not less than two coats of enamel, in lieu of top roller coated. The eyelets shall have an outside diameter of 0.385 to 0.390 inch and a minimum inside diameter of 0.180 inch.

3.3.18 Nails for heel attaching. The nails for heel attaching shall be brass cut or wire type and shall be of sufficient length when driven to the top surface of the core to produce a clinch on the insole. The cut type of nails shall be cut from material conforming to alloy No. 260 or 268 of ASTM B 36 and shall be commerical No. 450 or 1336 with a 20-gage point. The wire type of nails shall be commercial type 13-1/2 gage with a flat clinching point and shall be cut from material conforming to alloy No. 268 of ASTM B 134.

3.3.19 Tacks and staples. Tacks for the heel seat lasting, tacks for the welt butting, and tacks or staples for attaching the shank shall be brass, shall be of sufficient length so that the parts through which they are driven will be thoroughly attached, and shall leave the insoles smooth on the inside. The tacks shall be cut from material conforming to alloy No. 260 or 268 of ASTM B 36.

* 3.3.20 Rib fabric. The fabric for the preformed ribs shall be an unbleached cotton or cotton-synthetic fabric. The fabric for the preformed ribs shall weigh a minimum of 8 ounces per square yard, shall have a minimum of 50 yarns per inch in the warp and of 30 yarns in the filling, and a minimum breaking strength of 105 pounds in the warp and the filling.

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3.3.21 Pegs and nails for heel seat fastening. The pegs for heel seat fastening shall be glue sized paper-base fiber, with a diameter of 0.081 to 0.086 inch, commercial type B or equal. The pegs shall be of sufficient length so that they will firmly secure the parts through which they are driven and leave the insoles smooth on the inside. No. 39 head, steel loose nails in lieu of pegs may be used for the heel seat fastening. The nails shall be of sufficient length to clinch properly.

* 3.3.22 Wax for thread. The wax for the thread used for the inseaming and the Goodyear stitching operations shall be white or golden and shall be a permanently plasticized resin that, in the temperature of normal machine use, shall thoroughly wax the thread for attaching the welt and the soling materials.

3.3.23 Bottom filler. The bottom filler may be either a thermoplastic or a cold process type.

* 3.3.23.1 Thermoplastic type. The thermoplastic type of bottom filler shall consist of a mixture of ground cork and a suitable thermoplastic binder in proportion by volume of three parts of cork to one part of binder. The cork granules shall be the best quality and substantially free from the outer layer of bark. The ground cork and the binder shall be thoroughly and evenly mixed. The binder shall be water insoluble and flexible. The binder shall have a softening point of at least 125 degrees Fahrenheit and a maximum penetration of 85 millimeters with a 200-gram load for 60 seconds at 77 degrees Fahrenheit before combining.

3.3.23.2 Cold process type. The cold process type of bottom filler shall consist of a mixture of ground cork and a suitable binder in proportion by volume of a minimum of two and one-half parts of cork to one part of binder. The cork granules shall be of the best quality and substantially free from any part of the outer layer of bark. When the cold process type of spread filler is dry and set, it shall consist of approximately four parts of cork to one part of binder. The cold process type of filler shall be spreadable without the use of heat. The cold process type of filler shall not soften at less than 150 degrees Fahrenheit and shall be water resistant and flexible.

3.3.24 Adhesives

3.3.24.1 Upper seams and inseam areas. The adhesive for sealing the upper seams and the inseam areas shall be either a vinyl type or a Buna-N type of adhesive. The vinyl type of adhesive shall be composed of a polyvinyl butyral base and a nonbleeding pthalate plasticizer and any other ingredients required to produce a clear, colorless adhesive of the highest quality suitable for the purpose intended. The gallon weight of the vinyl type of adhesive shall be not less than 7.4 pounds with a minimum total solids content of 19 percent and a viscosity of 40 to 50 seconds. The Buna-N type of adhesive shall conform to MIL-S-4383 except that the adhesive shall be natural color or colorless.

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* 3.3.24.2 Outsoles and midsoles. Adhesives that are compatible with carboxylated acrilonitule rubber compounds and are resistant to jet fuels shall be used for bonding the midsoles to the outsoles and may be used for bonding any of the parts of the boots for which a type of adhesive is not specified herein; however any of the adhesives utilized shall be one of the following three types: synthetic rubber solvent cement, synthetic resin and rubber solvent cement, or synthetic resin solvent cement. The cement shall form a firm and suitable bond between the components of the boots and shall meet the specified minimum adhesive strength for the bonding of the midsoles to outsoles (see 6.7) and for the bonding of the welting to the outsoles before and after immersion of the bonded specimen. The minimum adhesive strength for bonding the welting to the midsoles shall be 50 psi.

3.3.25 Shank combination. The shank combination shall consist of a steel shank attached by four prongs or two rivets to a shank board cover so that it is 5/8 inch (plus or minus 1/16 inch) from the front edge of the shank board cover. The shank combinations used on the boots shall conform to the sizes specified in table V.

TABLE V. Shank Combination Sizes

Shank Combination	Boot Sizes		
	Width A	Widths B, C, D, and E	Widths EE and EEE
Size 1	6-1/2 - 8	6 - 7-1/2	7-1/2
Size 2	8-1/2 - 9-1/2	8 - 9	8 - 9
Size 3	10 - 11	9-1/2 - 11	9-1/2 - 11
Size 4	11-1/2 - 12	11-1/2 - 12-1/2	11-1/2 - 12

3.3.25.1 Shank board covers. The shank board covers shall be cut right and left and in five sizes for the run of the boots for size 6-1/2 through size 12-1/2 in the proper widths to fill the entire width of the bottom cavity from the ball line to and including the heel seat, except for clearance that may be incurred in the shank casing schedule specified in table VI. The finished shank board covers shall have a thickness of 6-1/2/64 to 9-1/2/64 inch, a minimum bursting strength of 30 pounds, and a graduated scarf of 1-1/2 inches (plus or minus 1/16 inch) on both ends. The shank board covers shall be permanently marked with the applicable size in accordance with the shank casing schedule, as shown in table VI, which consecutively assigns the figures 1 through 5 according to size, using the figure 1 for the smallest size and the figure 5 for the largest size.

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TABLE VI. Schedule of Shank Casing Sizes

Boot Sizes (All Specified Widths)	Shank Size
6 and 6-1/2	1
7 and 7-1/2	2
8, 8-1/2. 9, and 9-1/2	3
10, 10-1/2, and 11	4
11-1/2. 12, and 12-1/2	5

3.3.25.2 Steel shank. The steel shank shall be made from 19-gage, 0.0418 inch (plus or minus 0.003 inch) thick, cold-rolled carbon steel with hardness ranging from 45 to 54 Rockwell C scale. The steel shank shall be shaped to conform to the profile of the bottom of the Type FWU-5/P last. The steel shank shall be made with a single or a double rib, and the overall thickness including the steel shall be 0.08 to 0.125 inch. The rib shall taper to a flat 5/8 inch (plus or minus 1/8 inch) at each end. The width of the shank shall be 0.625 inch (plus 0.003 or minus 0.007 inch). A 1/8-inch (plus or minus 1/32-inch) hole shall be centered 1/2 inch (plus or minus 1/32 inch) from the heel end of the shank. The length of the steel shank shall be as specified in table VII.

TABLE VII. Length of Steel Shank

Shank Size	Length in Inches (±1/32)
1	4-3/8
2	4-5/8
3	4-7/8
4	5-1/4
5	5-5/8

* 3.3.26 Laces. The laces shall be black and shall conform to type II, class 1 of V-L-61. The laces shall be a minimum of 60 inches long.

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* 3.3.27 Cleaning solution. The cleaning solution shall be a composition that, when applied with a cloth or sponge, shall thoroughly remove grease and soil marks from the manufacturing operations employed on the uppers of the boots. The pH value of the solution shall not exceed eight and shall not decrease the water resistance characteristics of the upper leather below the requirements specified. The supplier shall furnish a certificate of compliance that the pH value of the solution does not exceed eight (see 6.2).

3.3.28 Repairers. The repairers may be a crayon or paste type applied by hand, a liquid type applied by hand, or a liquid type applied by a spray method. The repairers shall match the color of the upper leather and shall have sufficient coverage to correct minor surface imperfections or discolorations of the leather.

3.3.29 Fillers. Fillers, if used, shall provide a suitable foundation for the application of the top finish and may be applied by sponge or spray method.

3.3.30 Top finish. The top finish shall be a silicone (Syl-mer) or alkenyl succinic acid (Bavon) composition that will be compatible with the previously applied compounds (see 6.4), shall provide a flexible finish with a medium bright luster, and may be applied by either the sponge or spray method.

* 3.4 Design. The boots shall be a 9-inch-high, blucher style with water resistant seam welt construction, water resistant full grain upper, fold down top cuff, full grain leather lining, and vinyl welting. The boots shall have pyroxylin or polystyrene box toes and jet fuel resistant soles, heels, and midsoles. The boots shall be equipped with a cinch clamp to fasten the laces, with speed lacers, and with eyelets.

3.5 Construction

3.5.1. Cutting. The uppers, the lining, the gusset, and the top turn-down cuff shall be cut from the leather specified herein. The vamp shall be cut from the bend area of the side leather. Line marker patterns shall be used for all upper leather fitting. No die stab marking shall be permitted.

3.5.2 Skiving. The parts of the uppers shall be skived on the flesh side at the points specified in table VIII. The width of the scarf shall be sufficient to permit the correct lasting of the boots and the required stitching without hardness or bulkiness at the seams. The feather edge of the scarf shall be approximately one-third the thickness of the leather, and the width of the scarf shall be 3/16 inch (plus or minus 1/32 inch).

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TABLE VIII. Skiving of Uppers

Parts of Uppers	Location of Skiving
Quarters	Top edge, vamp margin, and speed lacer loop row
Vamp	Throat (gusset joining)
Backstay	Bottom
Linings	Vamp throat, quarter, top, bottom, and back
Top band-cuff	Bottom edge of band

3.5.3 Staining of cut edges. All cut leather edges that will be exposed in the end item shall be stained to match the color of the upper leather.

* 3.5.4 Crimping. The vamps and the linings shall be crimped, if needed, one pair at a time. They shall be placed accurately on the crimping blade, and the machine plates shall be properly heated and under carefully adjusted pressure. The crimping blade shall be shaped to conform to the boot front and the Type FWU-5/P lasts.

* 3.5.5 Marking. The contractor's name, the number and the date of the contract, the purchasing office, and the size and the width of the boot shall be legibly stamped with marking ink on the quarter lining. The first line of the marking shall be approximately 3/8 inch from the bottom edge of the top facing.

3.5.6 Cement lining. The leather lining shall be spot cemented to the flesh side of the gusset, the vamp, and the quarters.

* 3.5.7 Sewing operations (except inseaming of welt). Unless otherwise specified herein, the upper stitching shall be performed with stitch type 301 of FED-STD-751 and shall be not less than 8 nor more then 10 stitches per inch.

3.5.7.1 Join uppers. The combination top band and cuff shall be attached to the inside of the quarter with one row of stitching at the boot top 1/16 inch (plus or minus 1/32 inch) from the top edge of the boot. The end of the band shall extend from the inside edge of the lacer stay on one side, around the top of the boot to the inside edge of the lacer stay on the opposite side. The quarters shall be stitched to the vamp with two double rows of stitching, single or double needle, along the sides. The inside rows of the double rows of stitching shall be spaced 1/8 inch (plus 1/8 inch or minus 1/32 inch) apart. The outside row of the bottom double rows of stitching shall be 1/16 inch (plus

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or minus 1/32 inch) from the edge of the quarter. Prior to stitching, the skived edges of the vamps shall be coated with sufficient viscous seam sealer (see 3.3.24.1) to provide the required sealing action. The cemented areas on the quarters shall be joined to the corresponding areas on the vamps within a period of not more than 45 minutes.

* 3.5.7.2 Join lining. The quarter lining and the vamp lining shall be joined at each side with a single row of stitching, quarter over vamp, with a single row of stitching 1/16 inch (plus or minus 1/32 inch) from the edge. The quarter lining and the vamp lining may be joined by cementing prior to stitching. The quarter lining shall be joined at the back with a single row of stitching 1/16 inch from the edges. The seam shall be rubbed out thoroughly and reinforced with the back seam tape, using stitch type 301 and seam class SSf of FED-STD-751, 1/16 inch from each edge of the tape and using 9 stitches (plus or minus 1 stitch) per inch.

* 3.5.7.3 Join backstay to quarter. The backstay shall be stitched to the quarter with two rows of stitching, single or double needle, spaced 1/16 inch (plus or minus 1/64 inch) apart with the first row of stitching approximately 1/16 inch from the edge.

* 3.5.7.4 Join gusset lining to gusset. The gusset lining shall be stitched to the gusset with one row of stitching 1/16 inch (plus or minus 1/32 inch) from the edge of the lining. The lining shall extend to within 1/2 inch (plus or minus 1/4 inch) of the bottom of the gusset.

3.5.7.5 Join gusset to vamp. The completed gusset shall be stitched to the vamp with two rows of stitching. The rows of stitching shall be 1/4 to 3/8 inch apart with the row closer to the vamp 3/32 inch (plus or minus 1/32 inch) from the bottom edge of the gusset and not closer than 1/16 inch from the edge of the vamp throat edge. The two rows of stitching shall converge at a point under the blucher ear. The sides of the gusset shall be stitched through the combined quarter and lining with two rows of stitching 1/2-inch (plus or minus 1/16-inch) gage to form the speed lacer row.

* 3.5.7.6 Barring. The vamp and the quarter shall be barred with two rows of stitching, uniformly spaced, parallel to the top row of stitching of the vamp and quarter double stitching rows on each side of the boot. The barring shall be 5/8 inch (plus or minus 1/16 inch) in length on each blucher ear and shall be accomplished by using a single or double needle with a 1/16-inch gage or an automatic method with tying or locking ends. The barring shall start at a point 1/8 inch (plus or minus 1/32 inch) from the front edge of the quarter and continue across the inside stitching on the speed lacer row.

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3.5.7.7 Closing quarters and vamps. The quarters and the vamps shall be closed at the back, using seam type FSa-1 and stitch type 304, 404, or 501 of FED-STD-751. The seams shall be not less than 1/16 inch nor more than 3/32 inch from the edges and shall be carefully and thoroughly flattened by rubbing.

* 3.5.8 Installing speed lacers. Twelve speed lacers shall be installed on each boot in the placement indicated by the marker patterns. Each speed lacer shall be installed so that the loop will not be deformed and so that at least 1/8 inch of the loop will project past the edge of the quarter.

* 3.5.9 Lacing for lasting. The lacing for lasting shall be accomplished by lacing not less than two of the lower three lacing loops or loop holes. The machine lacing for lasting shall provide for 1-inch (plus or minus 1/8-inch) opening after lasting.

* 3.5.10 Coating and sealing vamp-quarter seam connection. The inside upper stitching or the bobbin threads at the vamp-quarter seam connection shall be coated with a light continuous coat of the adhesive specified in 3.3.24.1. No adhesive shall be applied to either the upper or the welt-vamp juncture on the finished boots.

3.5.11 Insoles. The insoles shall be sorted for varying fibers to obtain uniform channeling and cased for even weight. The insoles shall be lightly degrained or heavily buffed. The insoles shall be fleshed and rounded to the contour of the Type FWU-5/P lasts. Die-cut insoles will be accepted, provided they conform to the dimensions of the Type FWU-5/P lasts. A tolerance of plus or minus 1/64 inch from the last dimensions will be allowed.

3.5.11.1 Marking and scoring. Each insole shall be stamped, in the center of the shank on the grain side, with the correct size and width, using numbers for the whole and half sizes and letters for the widths as shown in the schedule of sizes in table I. The stamping shall be visible after the heel pads have been installed. The marking shall be at least 1/4 inch high, and the indentation shall be 1/32 inch deep. The lip scoring shall be located 2-9/16 inches (plus or minus 1/16 inch) from the heel end of a size 8 insole and shall grade from size 8 at the rate of grade produced from a lip cutting and scoring machine.

* 3.5.12 Ribs. The insoles may have either channeled ribs or preformed (stuck-on) ribs. The preformed ribs shall be not less than 5 irons nor more than 6 irons thick.

* 3.5.12.1 Channel ribs. If channeled ribs are used, the insoles shall be channeled on the flesh side. The channel leaf and the outside lip shall be not less than one-third the thickness of the insoles; and the channel leaf shall be slightly thicker than the lip. The substance between the channel

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leaf and the lip shall be 1/64 inch greater than the combined thickness of the leaf and the lips. The channel leaf shall provide a height of rib to meet the requirements of the finished rib specified in 3.5.12.1.1. The channel margins shall be based upon the dimensions necessary to provide the sole extensions specified in 3.5.24.1. The lips shall be cemented with the synthetic rubber solvent cement or the synthetic resin and rubber solvent cement specified in 3.3.24.2 and pressed together at lip setting to incline inward.

* 3.5.12.1.1 Install insole reinforcement. If the channeled ribs are used, the insole reinforcement shall be coated on one side with the adhesive specified in 3.3.24.2 and shall be bonded to the flesh side of the insoles. The insole reinforcement shall cover the entire area between the ribs, from the score line forward, extend to the top of the standing rib, and be firmly and smoothly fitted and bonded to the lip base. The insole reinforcement shall be trimmed to the rib height of 15/64 inch (plus or minus 1/16 inch). A continuous coat of the adhesive specified in 3.3.24.2 shall be applied to the flesh side of the insole prior to the attachment of the insole reinforcement. The 3/4-inch-wide tape specified in 3.3.12 for reinforcing the insole may be used as an alternate method.

* 3.5.12.2 Preformed ribs. If the preformed ribs are used, the finished rib shall be 15/64 inch (plus or minus 1/64 inch) high, not less than 5/8 inch wide measured from the inside vertical portion of the rib, and shall extend from heel breast line to heel breast line on the insole. The rib shall be made from coated fabric or a combination of coated fabric and fiberboard. The rib fabric shall conform to the requirements specified in 3.3.20. The upstanding portion of the rib shall be one of the following: at least three plies of coated fabric with a fiberboard filler not less than 0.05 inch thick or fiberboard end coated fabric. The rib fabric specified in 3.3.20 shall be coated on one side with a rubber-based adhesive specified in 3.3.24.2 and firmly and smoothly fitted and bonded to the flesh side of the insole to provide for the required edge extension on the finished boot. The fabric or the fiberboard shall cover 5/32 inch (plus 1/32 or minus 2/32 inch) of the peripheral edge and extend over the top of the vertical rib and cover the area between the ribs for a distance not less than 5/8 inch from the inside rib edge. The rib shall have a minimum shear strength of 70 pounds and a minimum stitch strength of 20 pounds.

3.5.13 Lasting. The lasts, the insoles, the counters, the lining, and the box toes shall be properly assembled in accordance with the size and width of the uppers. The method of mulling is optional except that the boots shall not be dipped in water. Insoles of the correct size and width shall be tacked to the last with not less than five tacks: one tack in the center of the heel seat, one tack at the shank, one tack on each side of the ball, and one tack at the toe. The tacks shall not be larger than 2-1/2 ounces. The edges of the insole shall be flush with the last at all points. The heel seat of the

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insole shall be smooth and even with the heel seat of the last. The proper size counter for the size boot that is being lasted shall be well cemented before insertion between the lining and the upper. The correct size box toe shall be properly conditioned and inserted between the lining and the vamp section of the upper. The uppers shall be assembled to the lasts to provide a wiped-in heel seat of 1/2 to 9/16 inch and drawn over the lasts with a strong tension on the pulling-over machine to assure that the quarters at the blucher points and the vamps are down to the wood and the blucher noses are even. The sides of the boots shall be spindled, drawing the upper snugly to the last and then side lasted so that, when stapled, the upper will be held firmly and securely to the last. Care shall be taken so that the vamp lining is pulled smooth and tight without tears. The heel seat shall be wiped-in flat and free from wrinkles. The toe shall be firmly and smoothly wiped-in and shall be securely attached around the toe at the base of the insole rib. Tacks shall not be used above the lasting line.

* 3.5.13.1 Time allowance on lasts. The boots shall remain on the lasts until the boots are thoroughly dry. In any case, the lasts shall not be pulled the same day the boots are lasted.

* 3.5.14 Inseaming and seam sealing. After the lasting, the pulling of the tacks, and the toe trimming, the entire bottom edge of the boots or the lasted-over margin including tacked or punctured edges and staples from breast to breast or slightly beyond the welt butts shall be coated with the seam seal adhesive specified in 3.3.24.1. The vinyl welting shall be seamed with the thread specified in 3.3.13.3. The thread shall be thoroughly hot waxed, with proper tension on the thread, using a needle not larger than No. 41 and with not less than 3-1/4 stitches per inch. The seam shall be tight (see 6.5), and the inseam shall be stitched from insole rib butt to rib butt extending to points which will catch in the counter with one or two stitches.

3.5.15 Tack pulling and inseam trimming. Care shall be taken so that all tacks have been removed and no broken tack points remain. The inseam shall be carefully trimmed closely and evenly from butt of welt to butt of welt without cutting or damaging the stitches. The end of the welt shall be skived with 5/8-inch (plus or minus 1/8-inch) bevel, and the butts shall be tacked.

* 3.5.16 Second seam sealing. The heel seat cavity, the feather line to the top of the rib, the inseam stitching, the channeling, the nail holes, the tack holes, and the staples shall be given a heavy coat of seam sealer and allowed to cure at room temperature.

* 3.5.17 Shank fitting and bottom filling. The shank shall be carefully selected for correct size in accordance with the casing schedule specified in table V. The shanks shall be inserted in position with the fiberboard cover filling the cavity between the edges of the vamp and the quarter from the

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ball line rearward to the back of the heel seat. The forward end of the shank shall be flush with the insole and fit the contour of the boot bottom back of the ball line. The shank shall be attached to the boot with two tacks or staples, one at each side of the steel piece at the rear end of the cover. The bottom filler shall be applied and firmly pressed into the insole cavity around the toe, extending to the forward end of the shank. The filler shall have a uniform and even surface, and the bottom shall present an overall flat smooth surface for the sole laying. Any cavities between the shank cover, the insole rib, and the heel portion shall be filled with the filler.

* 3.5.18 Midsole stock fitting. The inner surfaces of the midsoles and the outsoles shall be thoroughly cleaned and coated, over the entire surface, with the cement specified in 3.3.24.2. A coat of cement to serve as a primer shall be applied to the midsole. After the first coat of cement has been allowed to dry, a second coat of the same cement shall be applied to the midsole. A single coat of the same cement shall be applied to the buffed side of the outsole. The combined midsole and outsole shall then be molded to conform to the correct contour of the bottom of the boot. The minimum bond strength of the combined midsole and outsole shall be not less than 10 pounds per linear inch of width.

* 3.5.19 Sole laying. The bottoms of the boots shall be thoroughly coated with a continuous layer of the cement specified in 3.3.24.2. After the combined midsole and outsole are joined as specified in 3.5.18, they shall be laid on a sole laying machine with sufficient pressure exerted to assure good adhesion at all points. The soles shall be an adequate size and laid evenly to allow for the specified edge extension (see 3.5.24.1).

3.5.20 Rough rounding. The soles shall be smoothly rounded on a rough rounding machine to permit the edges to be trimmed square to the finished dimensions of the outside ball, the toe, and the inside ball when measured at right angles to the upper.

* 3.5.21 Goodyear stitching operations. The outsoles and the welts shall be stitched together with a lockstitch machine, using the running and the shuttle threads specified in 3.3.13.2. Both the running and the shuttle threads shall be waxed. A needle and an awl not larger than No. 45 shall be used. The stitching shall be 4-1/2 to 7 stitches per inch and lie on the surface of the welt and close to the outer edge of the welt on the finished boot. The lock shall be placed just under the surface of the outsole.

3.5.22 Heel seat fastening. The heel seat fastening shall be by the pegs or nails specified in 3.3.21. If the pegs are driven progressively, one at a time, the pegs shall be 4 per inch and spaced 1/8 inch (plus or minus 1/32 inch) from the edges of the insole. If the pegs are driven simultaneously, a total of 17 pegs shall be driven at the following locations: seven pegs grouped at

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the back of the insole, 9/32 inch apart; three pegs grouped at the welt butt area 9/32 inch apart on each side; and two pegs spaced between the two groupings on each side. These pegs shall be spaced 1/8 inch (plus or minus 3/64 inch) from the edge of the insole. If the nails are used, the nails shall be driven 3 per inch and spaced 1/8 inch (plus or minus 1/32 inch) from the edge of the insoles and extend from welt butt to welt butt.

* 3.5.23 Heel seat rounding. The midsoles and the outsoles shall be smoothly rounded in the heel seat area from butt of welt to butt of welt.

* 3.5.24 Heeling. Each heel shall be attached with 13 nails. The nails shall be of a length to insure a secure clinch on the insoles. Care shall be taken so that the heeling machine is equipped with proper length drivers to drive the heel nails firmly and evenly against the core so that the nail heads will sink below the rubber surface and will not be visible.

* 3.5.24.1 Edge trimming. The sole edges shall be trimmed square with an extension of 4/16 inch in the outside ball area and 3/16 inch in the toe and inside ball area, measured at right angle to the upper. The edges of the midsole, the outsole, and the welt shall be evenly joined to the heel material. A tolerance of plus or minus 1/32 inch on all specified edge trimming dimensions will be permitted.

3.5.25 Heel trimming and finishing. The edges of the heel shall be trimmed square (at a right angle in relation to the heel bottom) and scoured to a smooth finish free of indentations and ridges.

3.5.26 Finish

3.5.26.1 Preparation. The boots shall be properly cleaned so that accumulations of dirt, wax, cement, and any other foreign matter are removed.

* 3.5.26.2 Treeing. The boots shall be treed while on the lasts or on a treeing machine using right and left tree feet conforming to the Type FWU-5/P last. All wrinkles shall be removed. No materials that could damage the leather, the thread, or any other components shall be used on the boots.

* 3.5.26.3 Final finish. The boots shall be repaired, properly filled, and given a top finish using the sponge or spray method and the materials and methods specified in 3.3.27 through 3.3.30. All raw edges (including the backstay and the quarter) shall be stained to match the black upper leather.

3.5.26.4 Nails and tacks. Nails and tacks that have been left protruding through the insole and that cannot be removed shall be cut as close as possible to the surface with no protruding stumps left. A tack detector shall be used to detect any tacks left inside the boots.

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3.5.26.5 Heel pads. The heel pads shall be inserted grain side up. The heel pads shall be coated with an adhesive as specified in 3.3.24.2 over the entire surface of the flesh side and pressed firmly into the heel seat at all points.

* 3.5.26.6 Lacing and mating. The boots shall be properly mated. A lace shall be inserted through the top lacing loop of each boot and shall be looped and tied after insertion. A cinch clamp as specified in 3.3.16 shall be installed on each side of the laces prior to looping and tying.

3.5.27 Sole reinforcing. No reinforcing nails shall be used on the soles.

* 3.6 Instruction tag. The instruction tag shall be fabricated from commercial grade tag stock and shall be 2 inches by 3 inches. One corner of the instruction tag shall be perforated to receive the lace. The following information shall be printed, in black ink, on both sides of the instruction tag.

Clinch Clamp Installation and Lace Storage

1. Single knot lace at mid center prior to installing in boot.
2. Lace up boot; insert ends of lace through cinch clamp; knot ends of lace together.
3. Don boot; grasp lace ends, and pull up to tighten lacing.
4. Slide cinch clamp down lace to boot top and lock.
5. Pull up cuff; wrap lace around top of boot, pocketing lace between top of hook fastener tape and top of boot. Turn cuff down, and press cuff firmly to fasten hook and pile fastener tape.

3.7 Performance. The finished boots shall be water resistant.

3.8 Government loaned property

* 3.8.1 Patterns, markers, and dies. Standard sets of the component patterns and markers by widths, as listed in table IX, will be furnished by the procuring activity (see 6.2) and shall be used as the basis for making the manufacturer's patterns and dies. The manufacturer shall follow the patterns, except for the lasting allowance which may be determined by the manufacturer. The contractor shall furnish wood insole rounding patterns or dies to conform to the FWU-5/P last dimensions.

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TABLE IX. Patterns and Markers

Component	Sizes and Widths
Patterns	
Vamp	Whole and half sizes, all widths
Quarters	Whole and half sizes, all widths
Gusset and speed lacer row stay	Whole sizes, all widths
Top facing cuff	Three sizes, two widths
Backstay	Three sizes, all widths
Vamp lining	Whole sizes, all widths
Gusset lining	Three sizes
Quarter lining	Whole and half sizes, all widths
Pile and hook fastener tape	One size
Markers	
Vamp	Whole and half sizes, all widths
Quarters	Whole and half sizes, all widths

3.8.2 Lasts. The boots shall be made on the Type FWU-5/P lasts. All necessary sizes and widths of the Type FWU-5/P lasts will be loaned to the contractor by the procuring activity (see 6.2).

3.9 Workmanship. The boots shall be constructed in a thoroughly workmanlike manner.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection of the boots are classified as follows:

* (a) First article inspection

(b) Quality conformance inspection.

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* 4.3 First article inspection. If first article inspection is specified (see 3.1 and 6.2), the first article sample shall be inspected to determine compliance with all the requirements specified herein with respect to materials, design, construction, and workmanship. The first article sample shall consist of one or more finished pairs of the boots and a complete set of the materials and components used in the construction of boots. The first article sample shall have been manufactured by the supplier in the same facilities and with the same equipment that are to be used for the manufacture of the production items. The approval of the first article sample will authorize the commencement of production but shall not relieve the supplier of the responsibility for compliance with all of the requirements specified herein.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the following:

(a) In-process inspection

(b) End item inspection.

4.5 In-process inspection. Inspection, which cannot be performed in the finished boots, shall be performed during the manufacturing process to determine whether operations or assemblies are performed as specified herein. Nonconformance of any material or service to any specified requirement shall be cause for exclusion of the material or service from consideration for acceptance.

* 4.5.1 Testing of materials and components. In addition to the testing of materials and components specified in the documents referenced herein, materials and components shall be tested as specified in table X to determine conformance to the characteristics listed therein. The sample size shall be as specified in table XI. Failure of one or more of the sample units or lot average, if applicable, to meet the specified requirements shall be cause for rejection of the lot.

* 4.5.2 Testing of intermediate products

* 4.5.2.1 Assembled outsole-midsole combination. The assembled outsole-midsole combination shall be subjected to the dry and the wet bond strength tests described in 4.7.6.2 and 4.7.6.3 to determine conformance to the requirements specified in 3.5.18. The requirements shall be applicable to the sample unit. The sample unit for the dry and the wet bond strength tests shall be one outsole-midsole combination. The sample size for the dry and the wet bond strength tests shall be as specified in table XI. If the bond strength of one or more specimens or sample units is less than 10 pounds per linear inch of width or if the average bond strength of all the sample units is less than 12 pounds per linear inch of width, the lot represented shall be rejected.

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TABLE X. Testing of Materials and Components

	Material or Component, Unit of Product, and Characteristic	Test Method	Determinations Per Sample Unit	Sample Unit 1/
*	Upper leather, one side or skin (see 3.3.1 and 3.3.1.1)			8-inch-square piece
*	Material identification	Visual	1	
*	Chrome tanned	Visual	1	
*	Water resistance	8021 or 8131 of FED-STD-311 2/		
*	Break	ASTM D 2941	1	
*	Lining, one side or skin (see 3.3.2)			2-inch square piece
*	Material identification	Visual	1	
*	Chrome tanned	Visual	1	
*	Grain	Visual	1	
*	Finish	Visual	1	
*	Welting, 5 yards (see table II)			
	Hardness	ASTM D 2240 3/	3	
	Specific gravity	See 4.7.1.1	3	
	Brittle point	See 4.7.1.2	10	
	Tensile strength	Die E, ASTM D 412	5	5 yards

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TABLE X. Testing of Mateials and Components (Cont)

	Material or Component, Unit of Product, and Characteristic	Test Method	Determinations Per Sample Unit	Sample Unit 1/
	Elongation	Die E, ASTM D 412	5	
	Water absorption	See 4.7.1.3	3	
	Solvent resistance			
	Hardness	See 4.7.1.4.1		
	Tensile strength	See 4.7.1.4.2		
	Elongation	See 4.7.1.4.2		
	Volume swell	See 4.7.1.4.3	3	Ten 1/4-inch-long pieces
*	Polyethylene resin, 1 pair of counters (see table III)			
*	Specific gravity	ASTM D 792 (Method A)	1	One counter
*	Melt index	ASTM D 1238 4/		
*	Tensile strength	ASTM D 412 4/		
*	Elongation	ASTM D 412 4/		
*	Stiffness modulus (torsion) to 23 degrees centigrade	ASTM D 1043 4/		
*	Brittle temperature (80 percent nonfailure)	ASTM D 746 4/		

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TABLE X. Testing of Materials and Components (Cont)

	Material or Component, Unit of Product, and Characteristic	Test Method	Determinations Per Sample Unit	Sample Unit 1/
*	Water absorption, gain in weight in 24 hours	ASTM D 570 4/		
*	Volatility loss (72 hours at 82 degrees centigrade)	ASTM D 1203 4/		
*	Midsoles, 1 pair (see 3.3.10 and table IV)			
	Nonmarking	See 4.7.2.1	1	
	Hardness	3021 of FED-STD-601	3	Two pairs
	Cut growth after aging	See 4.7.2.2	2	
	Volume swell	See 4.7.2.3	3	
	Linen thread, 1 cone or spool (see 3.3.13.3)			
	Water absorption	See 4.7.3.1 and 4.7.3.2	1	One cone or spool
	Plastic box toes, 1 pair (see 3.3.15)			
	Material identification	Standard commercial	1	

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TABLE X. Testing of Materials and Components (Cont)

Materials or Components, Unit of Product, and Characteristic	Test Method	Determinations Per Sample Unit	Sample Unit 1/
Thickness	Standard commercial	3	One pair
Nails for heel attaching, 1 pound (see 3.3.18)			
Material identification	Standard commercial	1	
Finish	Standard commercial	1	1/4 pound
* Gaging of nail points	See 4.7.4	1	
Tacks and staples, 1 pound (see 3.3.19)			
Material identification	Standard commercial	1	
Finish	Standard commercial	1	1/4 pound
Bottom filler, 1 pound (see 3.3.23.1 and 3.3.23.2)			
Percentage of cork to binder	6341 of FED-STD-311	2	

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TABLE X. Testing of Materials and Components (Cont)

Material or Component, Unit of Product, and Characteristic	Test Method	Determinations Per Sample Unit	Sample Unit 1/
Solubility of binder in water	Standard commercial	1	
Flexibility of binder	Standard commercial	1	
Softening point of binder	ASTM E 28	2	
Penetration of binder	ASTM D 5	2	1-pound composite
Vinyl adhesive for seam sealing, 1 gallon (see 3.3.24.1)			
Composition	Standard commercial	2	
Gallon weight	ASTM D 816	1	
Total solids content	ASTM D 1084	2	One quart
Viscosity	ASTM D 1084	2	

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TABLE X. Testing of Materials and Components (Cont)

Material or Component, Unit of Product, and Characteristic	Test Method	Determinations Per Sample Unit	Sample Unit 1/
* Adhesive for outsoles and midsoles (see 3.3.24.2 and 6.7)			
Bonded strength of welting to midsole	Method B, ASTM D 816		One cemented unit 5/
Bonded strength of midsole to outsole	Method B, ASTM D 816	2 or 6 as applicable to test method	One cemented midsole-outsole combination
Shank combination, 1 pair (see 3.3.25.1 and 3.3.25.2)			
Shank board cover			
Material identification	Standard commercial	1	
Bursting strength of fiberboard	See 4.7.5	1	
Thickness of fiberboard	Standard commercial	3	One pair

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TABLE X. Testing of Materials and Components (Cont)

Material or Component, Unit of Product, and Characteristic	Test Method	Determinations Per Sample Unit	Sample Unit 1/
Steel shank			
Material identification	Standard commercial	1	
* Rockwell hardness scale C	ASTM E 18	1	
Thickness	Standard commercial	1	

* 1/ The requirement shall be applicable to the sample unit except for the adhesive, the shank combination and the bottom filler. The requirement shall be applicable to the lot for the adhesive, the shank combination, and the bottom filler.
* 2/ If method 8131 of FED-STD-311 is used, the location designated "Test Area a" on the figure entitled "Location of Test Area" in section 4 of FED-STD-311 shall withstand 4,000 flexes; however, 30 percent of the samples shall withstand 1,000 flexes prior to water penetration. If method 8021 of FED-STD-311 is used, the location designated "Test Area a" on the figure entitled "Location of Test Area" in section 4 of FED-STD-311 shall withstand 15,000 flexes; however, 30 percent of the samples shall withstand 3,000 flexes prior to water penetration.
* 3/ Fifteen-second reading on flat portion.
* 4/ A certificate of compliance will be acceptable for this requirement; however, the Government shall have the right to check test items to determine the validity of the certification.
* 5/ The cemented unit shall consist of six assemblies of welting strip joined to midsoles. The width of the test specimen shall be the same as the width of the welting. If failure occurs other than along the glue line, i. e., if the material fails before the full load is applied to the adhesive bond area, the test specimen will be considered to have complied with the requirement.

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TABLE XI. Sample Size

Lot Size	Sample Size
800 or less	2
801 to 22,000 inclusive	3
22,001 or more	5

* 4.5.2.2 Assembled insole. If the preformed (stuck-on) rib described in 3.5.12.2 is used, the assembled insole shall be subjected to the shear strength and the stitch strength tests described in 4.7.7.1 and 4.7.7.2 to determine conformance to the requirements specified in 3.5.12.2. The requirements shall be applicable to the sample unit. The sample unit for the shear and the stitch strength tests shall be one assembled insole. The sample size for the shear strength and the stitch strength tests shall be ten assembled insoles (five for each test) regardless of lot size. If the shear strength of one or more specimens or sample units is less than 70 pounds or if the average shear strength of all sample units tested is less than 75 pounds, the lot represented shall be rejected. If the stitch strength of one or more specimens or sample units is less than 20 pounds or if the average stitch strength of all the sample units tested is less than 30 pounds, the lot represented shall be rejected. The individual values utilized in expressing the final results shall be recorded in the test report.

4.5.3 Examination of materials and components. Each material and each component which cannot be checked after final assembly shall be examined in lots or batches, by sampling, for compliance with this specification and applicable and referenced documents; any deviation therefrom shall be classified as a defect. Unless otherwise specified, the applicable acceptance quality level (AQL) shall be 4.0 defects per 100 units (dhu).

4.5.3.1 Examination of insoles. The insoles shall be examined for the visual and dimensional characteristics specified in KK-I-570.

* 4.5.3.2 Examination of upper leather parts prior to fitting. The vamps and the quarters shall be examined for defects. The sample unit shall be one cut part. The lot size shall be expressed in the same terms as the sample unit. The inspection level for these examinations shall be I of MIL-STD-105. The AQL for the vamps shall be 2.5 dhu for major defects and 6.5 dhu for total defects. The AQL for the quarters shall be 2.5 dhu for major defects and 6.5 dhu for total defects. The defects found during these examinations of the vamps and the quarters shall be classified as specified in table XII.

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* TABLE XII. Examination of Uppers Prior to Fitting

Defect	Major	Minor
Vamp not full grain leather	X	
Weight of vamp more than 1/2 ounce less than minimum specified	X	
Weight of vamp more than maximum or up to 1/2 ounce less than minimum specified		X
Coarse rough fiber on flesh side of vamp	X	
Quarters not full grain leather	X	
Weight of quarters more than 1/2 ounce less than minimum specified	X	
Weight of quarters more than maximum or up to 1/2 ounce less than minimum specified		X
Coarse rough fiber on flesh side of quarters	X	

* 4.5.3.3 Examination of uppers after all fitting. The upper assembly shall be examined for defects in cutting, fitting, and any other construction characteristics which cannot be seen in the end item. The sample unit for this examination of the uppers after all fitting shall be one completely fabricated upper assembly prepared for lasting. The lot size for this examination of uppers after all fitting shall be expressed in the same terms as the sample unit. The inspection level for this examination of the uppers after all fitting shall be I of MIL-STD-105; the AQL shall be 2.5 dhu for major defects and 6.5 dhu for total defects. The defects found during this examination of the upper after all fitting shall be classified as specified in table XIII.

* 4.5.3.4 Examination of counters. If leather counters are used, the counters shall be examined in accordance with the visual examination and the dimensional examination specified for the end item in MIL-C-41814. If polyethylene counters are used, the counters shall be visually examination for conformance to the requirements specified in 3.3.7.2

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* TABLE XIII. Examination of Uppers After All Fitting

Defect	Major	Minor
Construction and workmanship		
Any component missing or not specified type	X	
Any component misplaced or not attached as specified, seriously affecting serviceability	X	
Any component misplaced or not attached as specified, affecting serviceability but not seriously		X
Vamp and quarter edge not seam sealed	X	
Construction not as specified, seriously affecting serviceability	X	
Construction not as specified, affecting serviceability but not seriously		X
Back seam not rubbed down		X
Back seam tape not caught in both rows of stitching		X
Leather		
Thickness of leather more than 1/2 ounce less than minimum specified	X	
Thickness of leather more than maximum specified or up to 1/2 ounce less than minimum specified		X
Coarse or rough fiber on flesh side of leather	X	
Leather containing slaughter cut or any other damage seriously affecting serviceability	X	
Leather containing slaughter cut or any other damage affecting serviceability but not seriously		X
Leather not full grain	X	
Leather not specified color	X	

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* 4.5.3.5 Examination of boot before bottom filling. The partially fabricated boot shall be examined for defects in construction and workmanship. The sample unit shall be one partially constructed boot, at a point after lasting and attachment of the shank but before the application of the bottom filler. The lot size shall be expressed in the same terms as the sample unit. The inspection level for this examination of the partially fabricated boot shall be I of MIL-STD-105, and the AQL shall be 2.5 dhu for major defects and 6.5 dhu for total defects. Defects found during this examination of the partially fabricated boot shall be classified as specified in table XIV.

4.6 End item inspection

* 4.6.1 Visual examination of end item. The end item shall be visually examined for defects of pairing, design, materials, construction, workmanship, finish, and marking. The inspection level for this visual examination of the end item shall be II of MIL-STD-105. The sample unit and the term for the expression of the lot size shall be one completely fabricated boot, except that the selection shall be by pairs and the pairs shall be examined together for pairing examination. Each defect found during the examination for pairing shall be classified as a single defect. Except for the critical defects specified in 4.6.1.1, the defects found during this visual examination of the end item shall be classified as specified in table XV. Except for the critical defects specified in 4.6.1.1, the AQL for this visual examination of the end item shall be 2.5 dhu for major defects and 6.5 dhu for total major and minor defects.

4.6.1.1 Critical defects. In addition to the examination specified in 4.6.1, the boots shall be examined for protruding points of tacks and nails. Any protruding point of a tack or a nail forward of the heel breast line shall be cause for rejection of the entire lot.

4.6.2 Testing of end item. The finished boots shall be tested for water resistance at the seam and inseam as specified in 4.7.8. The sample unit and the term for expression of the lot size for this testing of the end item shall be one pair of boots. The inspection level for this testing of the end item shall be S-2 of MIL-STD-105. If one or more sample units fail to pass this test, the lot represented shall be rejected.

4.6.3 Examination of preparation for delivery. The shipping container fully prepared for delivery shall be examined, for the defects listed in table XVI, to determine compliance with the requirements specified herein with respect to packaging, packing, and marking. The sample unit for this examination of preparation for delivery shall be one shipping container fully prepared for delivery except that it need not be sealed. The lot size shall be the number of shipping containers in the inspection lot. The inspection level shall be S-2 of MIL-STD-105, and the AQL shall be 2.5 dhu.

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TABLE XIV. Examination of Boots Before Bottom Filling

Defect	Major	Minor
Bottom of boot		
Any component missing or not specified type	X	
Shank improperly positioned		X
Shank not specified size or malformed		X
Shank not fitting contour of boot bottom	X	
Shank not securely attached		X
Any tear in insole reinforcement over 1/2 inch	X	
Any insole or anchor tack not removed	X	
Excess leather in heel seat interfering with proper fit of shank	X	
Upper damaged in lasting, seriously affecting serviceability	X	
Upper damaged in lasting, affecting serviceability but not seriously		X
Poor heel seat lasting, i.e., insufficient lasting tacks, or heel seat not flat	X	
Inseam not properly trimmed, seriously affecting serviceability	X	
Inseam stitch broken, cut, or damaged, e , stitch cut in trimming operation	X	
Inseam not properly trimmed, affecting serviceability but not seriously		X
Less than 3 stitches per inch on inseam	X	
Broken insole rib, inseam stitching not at bottom of insole rib, or stitching not in welt groove	X	

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* TABLE XIV. Examination of Boots Before Bottom Filling (Cont)

Defect	Major	Minor
End of counter not caught in inseam stitching		X
Welts not properly laid out		X
Ends of welt butts not skived or tacked		X
Inseam seal operation, before or after welting, omitted or not properly performed	X	
Any operation omitted or not properly performed seriously affecting serviceability	X	
Any operation not properly performed, affecting serviceability but not seriously		X
Upper part of boot		
Uppers not tightly lasted to wood		X
Plastic box toe out of alignment or crooked	X	

4.7 Test methods

4.7.1 Welting

4.7.1.1 Specific gravity. The specific gravity of the welting shall be determined by a Fisher Young gravitometer using a specimen that is 2 to 3 inches long.

4.7.1.2 Brittle point before and after aging. The brittle point of the welting shall be determined, before aging, in accordance with ASTM D 746. The specimen of the welting shall be aged by placing it in an oven and maintaining it at a temperature of 100 degrees (plus or minus 2 degrees) centigrades for seven days. After the 7-day period, the welting shall be removed and allowed to remain at room temperature for 24 hours. After the 24-hour period the welting shall be tested in accordance with ASTM D 746.

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* TABLE XV. Examination of End Item

Defect	Major	Minor
Pairing		
Boots not properly mated, i.e., not right and left of same size or noticeable variation in color or appearance	X	
Cleanliness		
Any spot, stain, or foreign matter clearly noticeable		X
Color and finish		
Boot not specified color or color not uniform	X	
Upper leather not finished to specified brightness		X
Finish streaky, chipped, or flaky on uppers		X
Any exposed cut of leather not stained to match uppers	X	
Design, type, and size		
Boot not specified design, type, or size	X	
Materials		
Any component not fabricated of specified material	X	
Stretchy leather in vamp	X	
Any shallow hole, marks, or any other open damage in leather	X	
Badly healed damage or scratch in leather, seriously affecting serviceability or appearance	X	
Any healed damage or scratch in leather, affecting serviceability or appearance but not seriously		X
Hole, abrasion, or any other damage in leather or leather otherwise inferior	X	
Leather not full grain	X	

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TABLE XV. Examination of End Item (Cont)

Defect	Major	Minor
Construction and workmanship		
Any component missing	X	
Any cut, tear, hole, rip, repair, abrasion, or any other type of factory damage seriously affecting serviceability or appearance	X	
Any component or assembly misplaced or operation omitted or not properly performed, seriously affecting serviceability or appearance	X	
Any component or assembly misplaced or operation omitted or not properly performed affecting serviceability or appearance but not seriously		X
Boot twisted or seriously misshapen	X	
Checked sole, i.e., separation of joined outsole and midsole or separation of sole and welting 1/		
Seams and stitching (upper)		
Any open seam 1/ 2/		
Loose tension resulting in a loosely secured seam	X	
Tight tension resulting in puckering or cutting of leather	X	
Not specified stitch or seam type	X	
Back seam containing less then 7 stitches per inch but not less than 5 stitches per inch		X
Back seam containing less then 5 stitches per inch	X	
Any row of stitching (except back seam and Goodyear stitching) with less than 8 stitches per inch but not less than 6 stitches per inch		X

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TABLE XV. Examination of End Item (Cont)

Defect	Major	Minor
* Any row of stitching (except back seam and Goodyear stitching) with less than 6 stitches per inch	X	
Stitches per inch (except Goodyear stitching) more than specified maximum stitches per inch but not damaging leather		X
Stitches per inch (except Goodyear stitching) more than specified maximum stitches per inch resulting in damage to leather	X	
Gage of stitching (except Goodyear stitching) not as specified or irregular seriously affecting serviceability or appearance	X	
Gage of stitching (except Goodyear stitching) not as specified or irregular affecting serviceability or appearance but not seriously		X
Required stitching omitted	X	
Thread ends not trimmed throughout boot		X
Needle hole or needle chew	X	
Stitching over-run resulting in stitching where not specified		X
Goodyear stitching (outsole)		
Goodyear stitching cutting into sole substance or deeper than specified	X	
Stitch channel deeper than specified, cutting outsole substance, or any Goodyear stitching not visible on surface of sole	X	
Less than 4 stitches per inch in Goodyear stitching	X	
* Less than 4-1/2 stitches per inch but not less than 4 stitches per inch or more than 7 stitches per inch in Goodyear stitching		X

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TABLE XV. Examination of End Item (Cont)

Defect	Major	Minor
Skipped or broken stitch in Goodyear stitching	X	
More than two stitches short in Goodyear stitching at heel breast	X	
One or two stitches short in Goodyear stitching at heel breast		X
Goodyear stitching running into traction pattern	X	
Gage of Goodyear stitching not as specified or irregular		X
Counter		
Rolled or curled counter	X	
Edge making		
Edge trimmed into Goodyear stitching	X	
Edge not trimmed square or trimming irregular		X
Sole extension more than 3/32 inch less than specified minimum	X	
Sole extension less than specified minimum but not more than 3/32 inch less than specified minimum		X
Sole extension beyond specified minimum		X
Heel finishing and attaching		
Heel not finished square, i.e., flared or tapered more than 1/8 inch		X
Heel not finished smooth, affecting appearance		X
Open heel seat	X	
Heel not specified size or type		X
Heel crooked, affecting serviceability		X

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TABLE XV. Examination of End Item (Cont)

Defect	Major	Minor
Checkered heel, separation of heel and outsole	X	
Eyelets		
Eyelets missing or insecurely clinched	X	
Edge of eyelet less than 1/16 inch or more than 5/32 inch from edge of quarter	X	
Thirteen nails in heels		
Any heel nail missing		X
Either corner breast nail not clinched on insole or clinched more than 3/16 inch	X	
Two or three nails (except corner breast nail) not clinched on insole		X
More than three nails (except corner breast nail) not clinched on insole	X	
Speed lacers		
Loops not properly spaced within row or misalignment between rows to an extent of interfering with proper lacing	X	
Loop projecting less than 4/32 inch past edge of quarter		X
Loop not securely clinched		X
Inseaming		
Grinning seam, i.e., thread exposed	X	
Strained seam, i.e., needle holes visible but thread not exposed		X
Insole		
Fiber pegs or loose nails not extending from welt butt to welt butt		X

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* TABLE XV. Examination of End Item (Cont)

Defect	Major	Minor
Insole short 3/16 inch or more	X	
Insole short 3/32 inch but less than 3/16 inch		X
Any protruding point of tack or nail in heel area	X	
More than two fiber pegs or loose nails run-off insole	X	
More than three fiber pegs or loose nails closer than 1/16 inch from edge of insole	X	
Lining		
Vamp lining wrinkled, excessive fullness, or loose, seriously affecting serviceability	X	
Vamp lining wrinkled, excessive fullness, or loose affecting serviceability but not seriously		X
Lining torn	X	
Marking		
Marking missing, incomplete, incorrect, not applied in specified manner, misplaced, illegible, or not specified size		X

1/ The defect shall be classified as a major defect if it seriously affects serviceability or appearance and as a minor defect if it affects serviceability or appearance but not seriously.
2/ A seam shall be classified as open if one or more stitches joining a seam are broken or if two or more continuous skipped stitches or run-offs occur. A seam shall be considered open if either one side or both sides of a double-stitched seam are open.

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TABLE XVI. Examination of Preparation for Delivery

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application
Materials	Any component missing or any component damaged, affecting serviceability
	Any component not as specified
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, or inadequate stapling
	Bulging or distortion of containers
Content	Number of pairs of boots per container more or less than specified
	Size shown on one or more pairs of boots not as specified on shipping container or package 1/

1/ Two pairs of boots from each container in the sample unit shall be examined for this defect.

4.7.1.3 Water absorption. A specimen of the welting shall be cut to a length of 4 inches (plus or minus 0.05 inch) and weighed to the nearest 0.001 gram, and the weight shall be recorded as W1. The weighed specimen shall be immersed in a beaker of distilled water and maintained for 7 days in an oven at a temperature of 70 degrees (plus or minus 2 degrees) centigrade. The beaker shall be removed from the oven and cooled to room temperature. The specimen shall then be washed in methanol, dried for 1 minute, and reweighed, and the weight shall be recorded as W2. The top of the beaker shall be sealed during this test to prevent the evaporation of the distilled water. The following formula shall be used to calculate the percent of water absorbed.

$$\text{Water absorption, percent} = \frac{W2 - W1}{W1} \times 100$$

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4.7.1.4 Solvent resistance

4.7.1.4.1 Hardness. The specimen of the welting shall be a minimum length of 3 inches of the full width and thickness of the welting. The specimen shall be completely immersed, at a temperature of 23 degrees (plus 2 degrees) centigrade for 24 hours (plus or minus 1/4 hour), in fluid conforming to type III of TT-S-735. The container shall be fitted with an air or reflux condenser so that a constant volume will be maintained. After the immersion period, the specimen shall be removed from the fluid, blotted dry with filter paper or any other absorbent material, and tested for hardness as specified in ASTM D 2240 with a 15-minute reading on flat portion.

* 4.7.1.4.2 Tensile strength and elongation. A specimen of the welting 6 inches by 0.45 inch by 0.125 inch (plus or minus 3 percent) shall be completely immersed, at a temperature of 23 degrees (plus or minus 2 degrees) centigrade for 24 hours (plus or minus 1/4 hour), in fluid conforming to type III of TT-S-735. The container shall be fitted with an air or reflux condenser so that a constant volume will be maintained. At the end of the 24-hour period of immersion, the specimen shall be removed from the fluid, blotted dry with filter paper or any other absorbent material, hung up in a temperature of 23 degrees (plus or minus 2 degrees) centigrade for 24 hours (plus or minus 1/4 hour), and then tested in accordance with ASTM D 412 (Die E) for elongation and tensile strength.

4.7.1.4.3 Volume swell. A 50-milliliter burette calibrated to 0.1 milliliter shall be filled with 30 milliliters of methanol which shall be recorded as VI. The test specimen shall be 1/4 inch (plus or minus 1/16 inch) in length. The test specimen shall be immersed in the methanol, and the increased volume shall be recorded as V2. The specimen shall then be removed, dried, and immersed in fluid conforming to type III of TT-S-735 for a period of 24 hours (plus or minus 1/4 hour). The treated specimen shall then be removed, dried, and inserted in the burette filled with 30 milliliters of methanol which shall be recorded as VIb, and the increased volume shall be recorded as V2b. The following formula shall be used to calculate the percent of volume swell.

$$\text{Volume swell, percent: } V2 - VI = VA$$
$$V2b - V1b = VB$$
$$\frac{VB - VA}{VA} \times 100$$

4.7.2 Midsoles

4.7.2.1 Nonmarking. One corner of the specimen shall be drawn firmly across a sheet of white bond paper. If the mark made can be easily erased by rubbing with the fingers, the specimen shall be considered to be nonmarking.

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4.7.2.2 Cut growth after aging. The midsoles shall be aged 24 hours at 100 degrees (plus or minus 1 degree) centigrade in accordance with method 7221 of FED-STD-601. After aging, the cut growth of the midsoles shall be determined in accordance with ASTM D 1052, except that the results shall be reported as percent of cut growth after 25,000 cycles and the reading after each 100 percent cut growth shall not be reported.

4.7.2.3 Volume swell. The volume swell of the midsoles shall be determined in accordance with method 6211 of FED-STD-601. The immersion medium shall be fluid conforming to type III of TT-S-735.

4.7.3 Water absorption of linen thread

4.7.3.1 Static absorption. A specimen of the unwaxed linen thread shall be tested in accordance with method 4502 of FED-STD-191, except that the specimen shall be tumbled for 2 hours.

4.7.3.2 Dynamic absorption. A specimen of the unwaxed linen thread shall be tested in accordance with method 4500 of FED-STD-191, except that the specimen shall be tumbled for 2 hours.

4.7.4 Gaging of nail points. The points of the nails used for heel attaching shall be gaged with an English standard wire gage in which 20 gage is 0.035 inch (plus or minus 0.005 inch). The points of the nails shall enter the gage at least 1/32 inch.

* 4.7.5 Bursting strength of shank board. The specimen, the full size of the shank board, shall be immersed in water for 2 hours at room temperature. The excess water shall then be removed by lightly wiping the specimen with an absorbent cloth or blotter. The specimen shall then be clamped into the test machine and tested for bursting strength. The test machine used to determine the bursting strength of the specimen shall be a Scott vertical tensile strength or Mullen type tester or any other acceptable method with an adapter consisting of a clamp with a 3/8-inch orifice and a round head plunger of 1/8-inch diameter. The speed of the plunger shall be 4 inches (plus or minus 2 inches) per minute. The pounds required to burst the specimen shall be reported as the bursting strength.

* 4.7.6 Strength of assembled outsole-midsole combination

4.7.6.1 Test apparatus. The test apparatus used to determine the bonded strength of the intermediate products shall be a power driven pendulum-type machine or an approved portable testing device of equal performance. The rate of travel of the power actuated grip shall be 2 inches per minute. The test apparatus shall be operated without any device for maintaining load indication.

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* 4.7.6.2 Procedure for dry bond strength. The specimen shall consist of an outsole-midsole combination that has been aged for at least 2 days. The outsole shall be separated from the midsoles for a distance approximately 2-1/2 inches from the toe end of the specimen. The separate ends of the specimen shall be clamped into the jaws of the test apparatus. The specimen shall extend outward at right angles to the direction of the application of the load. The test apparatus shall be started, and the specimen shall be pulled apart for a distance of 1 inch. At the instant the specimen is pulled apart 1 inch, the load indicated on the test apparatus shall be recorded. The load shall be divided by the width of the specimen at the corresponding line of separation.

* 4.7.6.3 Procedure for wet bond strength. The specimen shall be immersed in fluid for 24 hours (plus or minus 1/4 hour) at 73.4 degrees (plus or minus 3.6 degrees) Fahrenheit. The fluid shall conform to type III of TT-S-735. The specimen shall be allowed to dry for a 24-hour (plus or minus 1/2-hour) period under standard conditions which are a temperature of 73.5 degrees (plus or minus 2 degrees) Fahrenheit and a humidity of 50 percent (plus or minus 5 percent). After the specimen has been allowed to dry as specified herein, the outsole shall be separated from the midsole for a distance of approximately 2-1/2 inches from the toe end of the specimen.

* 4.7.7 Strength of assembled insole

* 4.7.7.1 Shear strength. A specimen that is 1 inch by 2-1/2 inches shall be cut from the insole. The long edge of the specimen shall be perpendicular to the rib and shall include an inch width of the upstanding rib and the edge of the insole. The upstanding rib shall be clamped in one jaw of a Scott or similar tester. The unattached leather shall be clamped to the bottom grip. The rate of travel of the grip shall be 12 inches (plus or minus 2 inches) per minute under no load. The bond separating or breaking force shall be recorded to the nearest pound. Specimen failures caused by rupture of the leather shall not be recorded.

* 4.7.7.2 Stitch strength. A specimen shall be cut as described in 4.7.7.1. Two holes shall be drilled, in the middle of the height of the upstanding rib, using a 5/64-inch-diameter drill. The holes shall be 1/4 inch (plus or minus 1/24 inch) apart equal distant from the center line that bisects the 1-inch dimension of the specimen. A steel wire that is 0.041 inch (plus or minus 0.0002 inch) in diameter and in a "U" shape or the largest "U" shape of a paper clip shall be passed through the two holes so that the clip will pull against the fabric side of the rib. The free leather end shall be clamped in the lower jaw while the wire ends are clamped in a wrapped top jaw. The specimen shall be pulled apart as specified in 4.7.7.1. The moment of initial tear shall be recorded as the stitch strength of the specimen.

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4.7.8 Water resistance (end item). The finished boots shall be tested for leakage at the seam and inseam by immersing the pair of boots in water, with water over the welting, for a 15-minute period. Any evidence of penetration of water on the insoles or the flesh side of the upper leather including any evidence of water under the sealer coat, during this period, shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging of the boots shall be level A or C as specified (see 6.2).

* 5.1.1 Level A. Each boot shall be wrapped separately in a sheet of paper. The paper shall be not less than 30 pound basis weight (24 by 36) and shall conform to grade B of UU-P-268. Each mated pair of boots shall then be packaged in a box that conforms to type CF, class domestic, style RSC, grade 125 of PPP-B-636. The outside dimensions of the box shall be approximately 14-7/8 inches long, 11-5/8 inches wide, and 5 inches deep. The box shall be completely sealed around the center length of the box with tape. The tape shall be 3 inches wide and shall conform to type III of PPP-T-45.

5.1.2 Level C. The boots shall be packaged to afford adequate protection against damage, during shipment from the supply source to the first receiving activity. The supplier may use his standard practice, provided it conforms to the requirements specified herein for level C packaging.

5.2 Packing. Packing of the boots shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A. Six pairs of boots of the same size and width, each packaged in a fiberboard box as specified, shall be packed in a shipping container that conforms to class weather-resistant, size 3 of MIL-B-17757. The six boxes of boots shall be placed in the shipping container so that the boxes will be on end (upright position) with two boxes along the length and three boxes along the width and with the marked ends of the boxes facing the top flaps of the shipping container for ready identification. Closure of the shipping container shall be as specified in the appendix of the shipping container specification. Towards the end of the contract or if the quantity of boots of the same size and width is less than the quantity of boots required per shipping container, mixed sizes (see 5.3.1) may be packed within the same shipping container.

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* 5.2.2 Level B. Six pairs of boots of the same size and width, each packaged in a fiberboard box as specified, shall be packed in a shipping container that conforms to class domestic, grade 275, size 3 of MIL-B-17757. The six boxes of boots shall be placed in the shipping container so that the boxes will be on end (upright position) with two boxes along the length and three boxes along the width and with the marked ends of the boxes facing the top flaps of the shipping container for ready identification. Closure of the shipping container shall be as specified in the appendix of the shipping container specification. Towards the end of the contract or if the quantity of boots of the same size and width is less than the quantity of boots required per shipping container, mixed sizes (see 5.3.1) may be packed within the same shipping container.

* 5.2.3 Level C. Boots packaged as specified shall be packed in a manner that will insure acceptance by the carrier and safe delivery at destination at the lowest transportation rate for such supplies. The shipping containers shall conform to the rules of carriers or the regulations applicable to the mode of transportation.

* 5.3 Marking. In additon to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

* 5.3.1 Mixed sizes. Each shipping container packed with mixed sizes (see 5.2.1 and 5.2.2) shall have white paper labels securely attached to the end and the side. The labels shall be 4 inches by 5 inches. The words "mixed sizes" and thereunder the quantity of pairs and the sizes of boots contained therein shall be legibly printed on the labels.

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6. NOTES

6.1 Intended use. The boots covered by this specification are intended to be worn for operational flight purposes by personnel of the Department of the Air Force.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Sizes and widths required (see 1.2)

* (c) Requirements for first article inspection (see 3.1 and 4.3)

(d) Specify requirement for certificate of compliance for cleaning solution (see 3.3.27)

(e) Provisions for furnishing standard sets of patterns and markers and Type FWU-5/P lasts which are covered by MIL-L-27885 (see 3.8.1 and 3.8.2)

(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Samples. Samples of the boots may be obtained from the procuring activity or as directed by the contracting officer.

* 6.4 Water-resistant treatments. The purpose of treating the leathers with water-resistant materials (see 3.3.30) is to bar water penetration under dynamic flex conditions such as walking in water puddles or in mud. Materials which impart water-resistant characteristics may be Syl-mer (silicone) and Bavon (alkenyl succinic acid), both of which are specific to leather. Syl-mer is manufactured by the Dow Corning Corp., Midland, Michigan. Bavon is manufactured by the Scientific Advance Inc., Columbus, Ohio.

* 6.5 Inseam. Warming of the vinyl welting just prior to inseaming can facilitate the tightness of the inseams (see 3.5.14).

* 6.6 Figures. Figures 2 and 3 show general design of the finished boots and are furnished for information only.

* 6.7 Midsole-outsole bond strength. It is recommended that a shear strength of 50 psi be obtained for the cement used for bonding the midsole to the outsole to assure compatibility of the cement to meet the midsole to outsole bonding requirements (see 3.3.24.2) before and after immersion when tested in accordance with ASTM D 816.

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Figure 2. Boot (Cuff Up)

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Figure 3. Boot (Cuff Fastened Down Over Stored Laces)

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* 6.8 Cinch clamp. The clinch clamp specified herein may be purchased from Gad-Jets, Inc., 3629 N. Dixie Drive, Dayton, Ohio 45414 or Western Tool and Manufacturing Co., Inc., 1940 S. Yellow Spring St., Springfield, Ohio 45506.

6.9 Changes. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This notation was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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